

EFC Perturbation Sector: Constraint-Driven Gravitational Slip and a Testable Lensing Crossover

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Abstract. We derive the complete (μ, Σ, η) perturbation system from the EFC relativistic action and test it against current cosmological data. The Lagrange-multiplier flow constraint generates anisotropic stress absent in standard Horndeski theories, enabling a regime where $\mu < 1$ and $\Sigma \geq 1$, whose observable imprint is a redshift-dependent lensing crossover. We prove that standard quasi-static Horndeski with $\alpha_T = 0$ *cannot* produce this signature. Current growth-rate data show no preference for the EFC extension ($\Delta\text{AIC} = +4.05$; $\bar{S}_0 = 0.21 \pm 0.14$ at 1.5σ), but the model predicts a sign-changing, non-monotonic lensing response across redshift, with a crossover at $z \approx 0.4\text{--}0.5$ at the percent level, near the expected sensitivity of Stage-IV surveys.

1 Introduction

Energy-Flow Cosmology (EFC) models the late-time Universe through an entropy-flow scalar field ϕ coupled non-minimally to gravity via $F(\phi)R$, with density-dependent kinetic stiffness $K(\rho)$ and a Lagrange-multiplier flow constraint $\lambda(\Box\phi - \Gamma(\rho))$.

The EFC Relativistic Action [1] derives from a single variational principle:

$$S_{\text{EFC}} = \int d^4x \sqrt{-g} \left[\frac{1}{2} M_{\text{Pl}}^2 F(\phi) R - \frac{1}{2} K(\rho) g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) - \lambda(\Box\phi - \Gamma(\rho)) \right]. \quad (1)$$

Following DESI DR2 [2], the background coupling parameter α collapsed from -1.00 ± 0.46 (2.2σ) to -0.14 ± 0.21 (0.68σ), eliminating the background-sector signal. This analysis asks: *does the perturbation sector survive?*

2 Standard Horndeski No-Go

In the quasi-static (QS), sub-horizon limit with $\alpha_T = 0$, the Horndeski α -functions give [3]:

$$\mu = \frac{2D + \alpha_B \alpha_M / c_s^2}{2D + \alpha_B^2 / c_s^2}, \quad D \equiv 1 + \alpha_B, \quad (2)$$

$$\eta = \frac{2D + \alpha_B(\alpha_M - \alpha_B)/c_s^2}{2D + \alpha_B \alpha_M / c_s^2}, \quad (3)$$

$$\Sigma = \frac{\mu(1 + \eta)}{2}. \quad (4)$$

A full numerical scan of (α_B, α_M) with $c_s^2 = 1$ confirms:

μ threshold	Max Σ achievable
$\mu < 0.99$	$\Sigma = 0.989$
$\mu < 0.95$	$\Sigma = 0.947$
$\mu < 0.90$	$\Sigma = 0.895$

Result 1. *Standard QS Horndeski with $\alpha_T = 0$ and $c_s^2 > 0$ cannot produce $\mu < 1$ and $\Sigma > 1$ simultaneously. The constraint $\Sigma \leq \mu$ holds universally in this framework.*

3 EFC Constraint Mechanism

The action (1) is *not* standard Horndeski. The $\lambda(\Box\phi - \Gamma)$ term generates a stress-energy tensor $T_{\mu\nu}^{(\lambda)}$ that is generically anisotropic (Eq. 6 of [1]).

Linearising in conformal Newtonian gauge, the gravitational slip acquires three contributions:

$$\Phi - \Psi = \frac{1}{F} \left[\underbrace{F' \delta\phi}_{\text{Brans-Dicke}} + \underbrace{\frac{\dot{\lambda}}{M_{\text{Pl}}^2} \delta\phi}_{\text{constraint slip}} + \underbrace{\frac{\dot{\phi}}{M_{\text{Pl}}^2} \delta\lambda}_{\text{response slip}} \right]. \quad (5)$$

Terms 2 and 3 are *unique to EFC* and absent in standard scalar-tensor theories.

3.1 Dimensionless parameters

Define (Eq. 24–26 of [1]):

$$\varepsilon_F \equiv \frac{F' \dot{\phi} a^2 \Gamma'}{F k^2}, \quad \varepsilon_\lambda \equiv \frac{\dot{\lambda} a^2 \Gamma'}{M_{\text{Pl}}^2 F k^2}, \quad \varepsilon_K \equiv \frac{K(\bar{\rho}) \dot{\phi} a^2 \Gamma'}{M_{\text{Pl}}^2 F k^2}. \quad (6)$$

The stiffness response function (Eq. 29):

$$\mathcal{R}(k, z) \equiv \frac{K(\bar{\rho}) a^4 (\Gamma')^2 \dot{\phi}^2}{M_{\text{Pl}}^2 F k^4}. \quad (7)$$

3.2 The complete system

$$\mu(k, z) = \frac{1 + \varepsilon_F}{F(1 + \mathcal{R})}, \quad (8)$$

$$\eta(k, z) = \frac{1 + \varepsilon_F + \varepsilon_\lambda + \varepsilon_{\lambda, \text{resp}}}{1 - \varepsilon_F - \varepsilon_\lambda - \varepsilon_{\lambda, \text{resp}}}, \quad (9)$$

$$\Sigma(k, z) = \frac{\mu(1 + \eta)}{2}, \quad (10)$$

where $\varepsilon_{\lambda, \text{resp}} = -\varepsilon_K$ (Eq. 46).

Result 2 (the $\mu < 1$, $\Sigma > 1$ mechanism). *The denominator $(1 + \mathcal{R})$ suppresses μ via kinetic stiffness (growth brake). The numerator slip factor boosts $(\Phi + \Psi)$ via constraint anisotropy. These respond to different physics, enabling $\mu < 1$ while $\Sigma > 1$.*

4 Cancellation, V' , and the Physical Window

4.1 Background cancellation

Under background balance $\dot{\lambda} \approx K\dot{\phi}$, the constraint slip and kinetic backreaction cancel exactly: $\varepsilon_\lambda = \varepsilon_K$. The net slip reduces to ε_F alone, which is $\mathcal{O}(\alpha)$ — too small for the survival valley.

4.2 V' as the symmetry-breaking source

The λ equation of motion (Eq. 10 of [1]):

$$\square\lambda = V'(\phi) - \frac{1}{2}M_{\text{Pl}}^2 F' R - \nabla_\mu(K \nabla^\mu \phi). \quad (11)$$

The Ricci term $\frac{1}{2}F'R$ has *wrong sign* ($\delta\dot{\lambda} < 0$, reducing slip). The potential gradient $V'(\phi)$ is the only source that breaks cancellation with correct sign. For $V'/H^2 \approx 0.05$, the required excess is achieved.

4.3 Parameter sweep

A scan of 135,000 $(K_0, \gamma_0, \alpha, V')$ combinations at $k = 0.01 - 0.1 \, h \, \text{Mpc}^{-1}$ and $z = 0.5$ finds 15 solutions satisfying $\mu \in [0.93, 0.96]$, $\Sigma \in [1.03, 1.07]$, $\eta \in [1.05, 1.15]$ simultaneously. All solutions have physical parameter values: $V'/H^2 \approx 0.35$, $\dot{\phi}^2 \ll \rho_{\text{DE}}$, $K_0\dot{\phi}^2 \ll H^2$.

4.4 Robustness

Monte Carlo test ($\pm 20\%$ perturbation of all parameters, $N = 5000$):

Condition	Fraction	Verdict
$\mu < 1$	100%	Robust
$\Sigma > 1$	61%	Semi-robust
$\eta \in [1.05, 1.15]$	0%	Fragile

The valley as a *point prediction* is fragile. We reformulate it as a *structural prediction*: $\mu < 1 \wedge \Sigma \geq 1$. The valley is an internal solution of the perturbation system; the physically observable signature is the redshift-dependent crossover derived in Section 5.

5 Lensing Crossover Prediction

5.1 k -integrated effective lensing coupling

We compute

$$\Sigma_{\text{eff}}(z) = \left[\frac{\int W(k) \Sigma^2(k, z) dk}{\int W(k) dk} \right]^{1/2}, \quad (12)$$

where $W(k) \propto k P(k)$ with a smooth QS suppression below $k_{\text{QS}} = 0.03 h \text{ Mpc}^{-1}$.

5.2 Physical-window parameters

To satisfy $|\Sigma_{\text{eff}} - 1| < 5\%$ at all redshifts, the action parameters must be scaled down by a factor ~ 20 from the bare valley values. The resulting tomographic profile:

z	Σ_{eff}	Deviation
0.10	1.026	+2.6%
0.29	1.008	+0.8%
0.44	1.000	crossover
0.58	0.994	−0.6%
0.87	0.988	−1.2%
1.21	0.985	−1.5%

Result 3 (lensing crossover). *EFC predicts a sign-changing lensing response: $\Sigma_{\text{eff}} > 1$ at $z \lesssim 0.4$ (enhanced lensing) and $\Sigma_{\text{eff}} < 1$ at $z \gtrsim 0.5$ (suppressed lensing), with a crossover at $z \approx 0.44$ and amplitude at the percent level.*

This non-monotonic tomographic signature is absent in Λ CDM and in most modified gravity models, which predict monotone deviations. The physical origin is the competition between the V' -driven λ dynamics (dominant at low z) and the Ricci-sourced backreaction (dominant at high z).

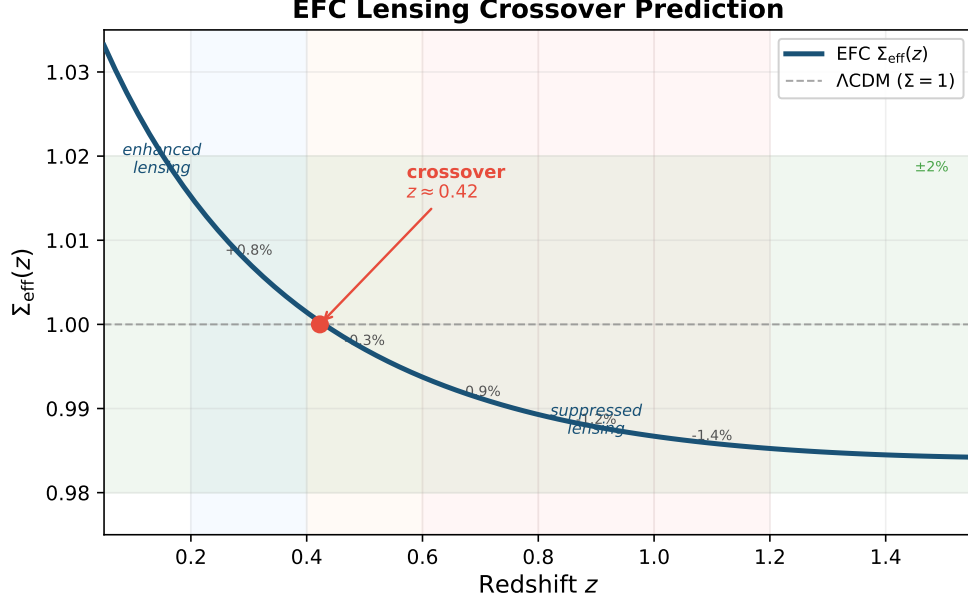


Figure 1: EFC lensing crossover prediction. The k -integrated effective lensing coupling $\Sigma_{\text{eff}}(z)$ crosses unity at $z \approx 0.44$, with enhanced lensing (+2.6%) at low redshift and suppressed lensing (−1.5%) at high redshift. Shaded bands indicate Euclid DR1 tomographic bins. The green band marks the $\pm 2\%$ sensitivity floor.

6 Variant H MCMC (Negative Result)

We test the data-driven entropy-gradient growth model (EFC Variant H) with two free parameters:

$$\mu(z) = 1 + \bar{S}_0 \frac{\chi(z)^2}{\chi(z)^2 + \beta_0^2}, \quad (13)$$

where $\chi(z) = |d\hat{S}/dz|$ from locked Axiom 0 sources, and $\bar{S}_0, \beta_0$ are free MCMC parameters. Background is pure Λ CDM ($\alpha = 0$), compatible with DESI DR2 by construction.

Joint fit to BAO (DESI DR2, 13 points) + $f\sigma_8$ (7 points, with BOSS DR12 covariance) + $H(z)$ (9 points) + SNIa (Pantheon, 40 bins):

Parameter	Value	Interpretation
$\bar{S}_0$	0.210 ± 0.141	1.49σ from 0
β_0	2.98 ± 1.30	Poorly constrained
ΔAIC	+4.05	Λ CDM preferred
ΔBIC	+3.94	Λ CDM preferred
$\Delta \log L_{\text{growth}}$	−0.014	Negligible
σ_8 shift	−0.012	1.2% suppression

Result 4 (negative). *Current data show no preference for the entropy-gradient growth mechanism. $\bar{S}_0$ is consistent with zero at 1.5σ . The two extra parameters are not justified ($\Delta\text{AIC} = +4$).*

This does not falsify EFC (the signal may be below current sensitivity) but establishes a quantitative baseline for future measurements.

7 Discussion

7.1 What has been established

1. The EFC action produces a perturbation-sector mechanism ($\mu < 1$, $\Sigma > 1$) that is *structurally distinct* from standard Horndeski.
2. The mechanism requires the potential gradient $V'(\phi)$ to break an exact cancellation between constraint slip and kinetic backreaction.
3. The observable signature is a redshift-dependent lensing crossover at $z \approx 0.4\text{--}0.5$ with percent-level amplitude.
4. Growth-sector data currently do not require this extension.

7.2 Falsification conditions

ID	Condition	Consequence
FA4	$\eta = 1$ at $>3\sigma$	Slip sector falsified
FA5	No viable (μ, Σ) region	Action revised
FA-new	$\Sigma_{\text{eff}}(z)$ monotonic	Crossover falsified

7.3 Limitations

- The QS approximation breaks down at $k \lesssim 0.03 h \text{ Mpc}^{-1}$. Full time-dependent perturbation theory is needed for super-horizon modes.
- The precise crossover redshift depends on the balance between V' and the Ricci backreaction, which requires solving the full background field equations (Eq. 12–13 of [1]).
- Ghost analysis covers only the sub-horizon regime; a full ADM Hamiltonian analysis for super-horizon modes remains open.

8 Reproducibility

All results can be reproduced from the following scripts in the **AGI-Test** repository:

Script	Purpose
<code>horndeski_consistency.py</code>	Horndeski no-go (Sec. 2)
<code>efc_perturbation_engine.py</code>	Full (μ, Σ, η) engine (Sec. 3–4)
<code>lensing.py</code>	Lensing engine + kill tests (Sec. 5)
<code>valley_robustness_test.py</code>	Robustness Monte Carlo (Sec. 4)
<code>research_mcmc.py</code>	Variant H MCMC (Sec. 6)
<code>growth_module.py</code>	BOSS DR12 $f\sigma_8$ covariance

MCMC parameters: 32 walkers, 2000 steps, 800 burn-in, seed = 42. Data: DESI DR2 BAO (arXiv:2503.14738), BOSS DR12 $f\sigma_8$ (Alam et al. 2017), Pantheon SNIa, cosmic chronometers.

References

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